Graphical Abstract

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Highlights

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Abstract

Approximately 600 million people in sub-Saharan Africa lack electricity access. Solar mini-grids offer practical solutions, but financing remains a major barrier. Carbon markets have been proposed as a way to generate additional revenue by selling carbon credits from emission reductions. However, little evidence exists on whether this approach actually works for small-scale rural energy projects. This study examines two questions: How effectively does carbon finance improve mini-grid investment and commercial viability, and what conditions are necessary for successful integration at scale? Through nine interviews with developers, carbon experts, policy officials, and registry professionals, plus focus groups with 30 community members across three Nigerian sites, the research reveals that carbon finance is supplementary rather than transformative. Developers did not base their investment decisions on carbon revenue. For projects under 200 kW, certification costs represent 8% of capital expenditure, while current prices of $1-5 per credit mean transaction costs exceed potential revenues. The atmosfair–MiniGridCo programme shows that combining multiple mini-grids into

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a single carbon project is technically possible when institutional support is available. However, the fact that it has not been replicated across more than 120 other Nigerian mini-grids points to a market failure. The conditions that made it work are not accessible to most developers. Successful integration requires coordinated public intervention across five areas: technical capacity, financial viability, regulatory frameworks, institutional arrangements, and carbon market design. Without structural transformation including publicly-funded aggregation and regulatory reforms, carbon markets remain largely inaccessible to mini-grids.

Keywords:

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1. Introduction

1.1. Background

Approximately 600 million people in sub-Saharan Africa (SSA) lack access to electricity, [1] representing more than half of the global population without power. This energy access deficit creates significant barriers to socio-economic development, exacerbates environmental degradation, and contributes to climate change. [2, 3]

Energy poverty in rural SSA manifests through multiple interconnected dimensions. Health complications arise from indoor air pollution caused by reliance on traditional biomass for cooking and kerosene for lighting. [4] Economic opportunities remain constrained by inadequate infrastructure to support income-generating activities and small businesses. [5] Educational outcomes suffer as children lack reliable lighting for evening study, while limited access to communication technologies perpetuates cycles of poverty and underdevelopment. [6]

Solar mini-grids have emerged as technically viable decentralized solutions for communities too remote for cost-effective grid connection yet sufficiently populated to support standalone commercial operations. [7] Solar mini-grids effect significant socio-economic benefits in recipient communities, [8] and the United Nations estimates decentralized renewable energy systems will provide clean energy access to two-thirds of rural SSA populations. [9] Despite this potential, mini-grid deployment remains constrained by high upfront capital costs, limited long-term financing, revenue uncertainty from rural consumers' constrained ability to pay, regulatory ambiguities, and high transaction costs relative to project size. [10, 11, 3]

Carbon markets have been proposed as supplementary financing mechanisms to address these constraints. By generating revenue from verified emission reductions, carbon credits could theoretically improve mini-grid commercial viability while simultaneously advancing climate mitigation objectives. [12] However, post-Paris governance fragmentation, with competing standards, divergent accounting rules, and uncertain host government authorization under Article 6, has complicated implementation. [13] Recent events underscore these risks: KOKO Networks, serving 1.5 million Kenyan households, entered administration in February 2026 after the Kenyan government refused a Letter of Authorisation, illustrating the extreme fragility of business models dependent on carbon revenues. [14] Whether carbon markets can meaningfully contribute to scaling rural electrification in SSA requires empirical examination.

1.2. Problem Statement

Significant knowledge gaps exist regarding the practical implementation and effectiveness of carbon finance for mini-grid development. Current literature provides limited empirical evidence on how carbon finance mechanisms perform in rural African contexts, what conditions enable successful integration, and how transaction costs compare to potential revenues at sub-200-kW scales. The complex regulatory landscape, including host government authorization requirements, [15, 13] revenue leakage to intermediaries, [16] and undefined treatment of carbon revenues within tariff frameworks, [17] creates uncertainty for developers and investors. Without clear understanding of these conditions, the potential of carbon markets to scale rural electrification remains largely theoretical rather than practical.

1.3. Research Questions

This research examines the potential of carbon markets in scaling up rural solar mini-grids in SSA addressing two questions:

RQ1: How effectively do carbon finance mechanisms incentivize investment and improve commercial viability of rural solar mini-grid projects?

RQ2: What technical, financial, regulatory, and policy conditions are necessary for successful carbon market integration at scale?

1.4. Significance and Scope

This research contributes empirical evidence on carbon market integration with rural electrification, advancing understanding of how market-based mechanisms can address development financing challenges. The findings inform policy frameworks for carbon market regulation, rural electrification strategies, actionable insights for developers, investors, and policy makers navigating post-Paris governance complexity. For developers, this research provides insights on project design and financing structures that enhance commercial viability through carbon market participation.

The research focuses on rural solar mini-grids in SSA, examining integration with voluntary carbon markets across East and West African contexts (i.e. Kenya and Nigeria) to capture variation in regional maturity and deployment models. Projects spanning small ($<$100 kWp), medium (100–400 kWp), and large ($>$1000 kWp) scales are examined to assess how carbon market viability varies by project size. The temporal scope focuses on developments from 2015 onwards, with particular attention to post-Paris Agreement development.

The findings are primarily applicable to SSA countries with foundational regulatory frameworks, adequate solar resources, and relatively stable political environments. Applicability beyond SSA depends on the presence of similar socio-economic, institutional, and development conditions.

2. Literature Review

2.1. Governance Fragmentation Under the Paris Agreement

International carbon markets have gone through boom, collapse, and post-Paris reconfiguration [18]. The Paris Agreement shifted from top-down targets to bottom-up nationally determined contributions (NDCs). Article 6 creates two pathways. Article 6.2 allows bilateral transfers of "internationally transferred mitigation outcomes" (ITMOs). Article 6.4 establishes a centrally governed crediting mechanism [13].

Ahonen et al. [13] call the current landscape "fragmented governance." Multiple public and private standards compete across market segments. This fragmentation hurts small-scale SSA projects most. Developers navigate competing standards (e.g., Gold Standard, Verra), divergent accounting rules, and uncertain demand. Host countries now bear "an unprecedented task of ensuring the environmental integrity and robust accounting of ITMOs." Many SSA governments lack this capacity.

2.2. *Environmental Integrity Risks*

sec:lit-risks) Schneider and La Hoz Theuer [15] define environmental integrity for carbon markets: transfers must lead to the same or lower global emissions. They identify four factors: robust accounting, unit quality, NDC ambition, and future mitigation incentives. Unit quality means one transferred unit represents at least 1 $tCO_2e$of actual reduction. Quality matters most when emission sources fall outside NDC scope or when a country's NDC requires no further mitigation. Both conditions apply to many SSA contexts.

There also exists a "hot air" risk: surplus units without real mitigation. La Hoz Theuer et al. [19] compared NDC targets with business-as-usual emissions across 131 countries. Hot air volume equalled total pledged mitigation. In their low mitigation scenario, hot air reaches 5.4 $GtCO_2e$. Between 36 and 52 percent of assessed NDCs could contain hot air.

Single-year NDC targets are common in SSA. They create accounting risks. Siemons and Schneider [20] compared two approaches: averaging counts credits in the target year based on multi-year average transfers and multi-year approaches using emissions trajectories. Averaging is problematic. Year-on-year emission fluctuations magnify ITMO volumes needed, creating uncertainty for developers. Multi-year approaches are more robust but require technical capacity to establish credible trajectories.

2.3. *Proposed Solutions and Their Limits*

Scholars have proposed technical fixes. Michaelowa et al. [21] introduced "ambition coefficients," declining multipliers applied to business-as-usual baselines, aligning crediting with net-zero pathways while respecting CBDR-RC. Developed countries would stop generating avoidance credits by 2035–2040, and developing countries could continue until 2050–2070. However, the approach needs regulatory coordination, technical capacity, and political consensus that most SSA contexts lack. The proposal also offers little guidance for small-scale commercial projects.

The World Bank has tried aggregation through carbon funds, alternating between "agenda setter" and "regime follower." Unfortunately, in the post-Paris period, its activities are "smaller, and...less transparent, than during the Kyoto Protocol period" [22]. This raises concerns about credit quality and replicability. Our empirical findings on the atmosfair-MiniGridCo model confirm these concerns.

2.4. Carbon Market Experience of Mini-Grids in Sub-Saharan Africa

The CDM's methodology AMS-III.BB, "Electrification of communities through grid extension or construction of new mini-grids," was adopted in 2012 and has since been updated twice [23]. Version 3.0 requires continuous metering and user surveys, caps annual emission reductions at 60,000 tCO_2e, and requires at least 75 percent of users must be households. This methodology provided the technical foundation for later mini-grid carbon projects under Gold Standard, Verra, and the Renewvia Environmental Exchange, including the atmosfair-MiniGridCo Programme of Activities in Nigeria.

Michaelowa et al. [24] provide the most comprehensive assessment of SSA carbon markets, identifying 457 private-sector mitigation projects. They note that SSA was "essentially bypassed by the CDM until the late 2000s," but Programmes of Activities (PoAs) increased participation. The SSA share in PoAs was ten times higher than for single projects. More recently however, implementation has stalled. Only 22 percent of single CDM projects and 16 percent of PoA component activities have issued credits. "Some may have paused or stopped operation due to the low market price of CERs."

Revenue leakage makes the problem worse. A Power Africa white paper found that only 30 percent of carbon credit revenues reach project developers, with the rest going to intermediaries, certification fees, and transaction costs [16]. More recently, SSA governments have demanded an additional cut of the proceeds. For example, Kenya's Climate Change Act added a 25 percent tax on carbon projects (40 percent for land-based projects) [25].

Ethiopian PoAs show both potential and limits. They enabled better CDM access but remain "restricted to sustainable energy access in rural areas due to the prohibitively low emissions baseline for grid electricity" [24]. South Africa's REIPPPP succeeded but is not replicable: "South Africa is particularly well positioned...As other SSA countries often still lack these capacities, REIPPPP cannot be easily replicated" [24].

Kenya illustrates the gap between theoretical potential and practical implementation. The country has a Climate Change Act (2016, amended 2023), Carbon Markets Regulations (2024), and a National Carbon Registry. Yet it has approved zero Article 6 projects [26]. The government fixed Corresponding Adjustment prices at USD 4.00 per $tCO_2e$and strictly limits Letters of Authorisation to preserve its "carbon budget" [27].

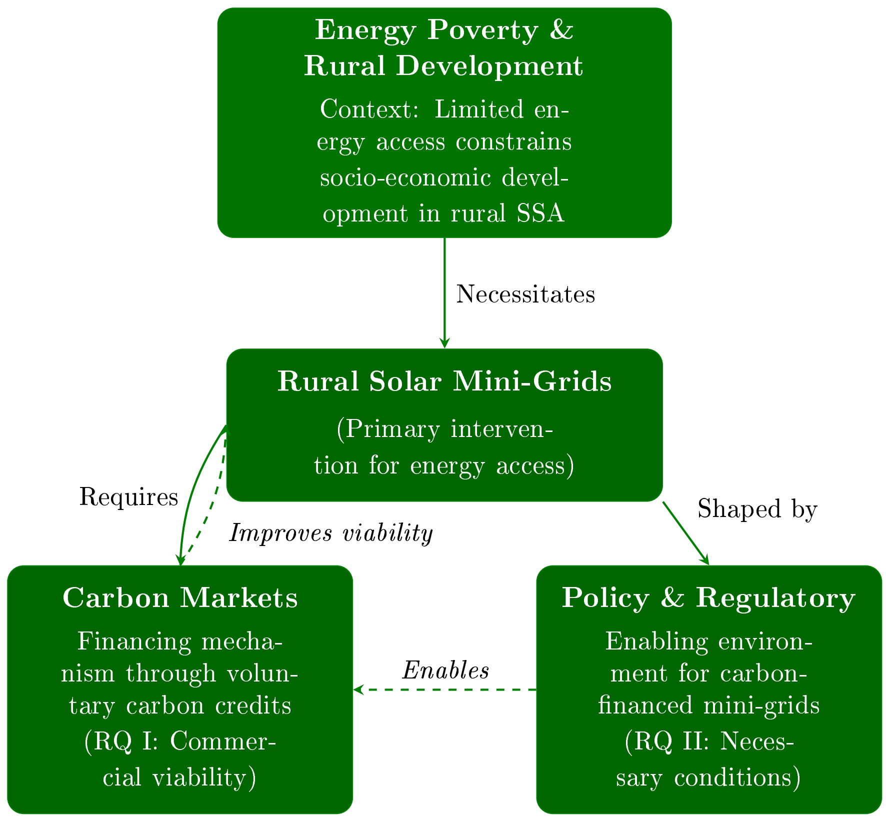
This gap led to the infamous collapse of KOKO Networks in February 2026. The company served 1.5 million Kenyan households with bioethanol cooking fuel, depended on carbon revenues to subsidize fuel prices. The government refused a Letter of Authorisation, barring monetization under Article 6.2 or CORSIA. KOKO entered administration. Seven hundred employees lost jobs, and 1.5 million households returned to charcoal and kerosene ([14, 28]. One analyst said this "highlights the extreme fragility of business models that rely on the complex and often shifting regulatory landscapes of international carbon markets" [28]. The case validates Schneider and La Hoz Theuer's [15] theoretical risk. Even real, additional, verifiable reductions cannot succeed without host government authorization in the Article 6.2 markets.

2.5. Research Gap and Contribution

The literature establishes three insights. First, carbon markets face integrity risks: hot air, additionality failure, accounting complexity. These threaten to increase global emissions [15, 19, 20]. Second, proposed governance solutions assume institutional capacities most SSA contexts lack [21, 22]. Third, evidence shows that while PoAs increased SSA participation, most projects remain stalled. Low prices, revenue leakage, and host government authorization barriers all play a role [24, 16, 14].

No empirical study has systematically examined how these risks and solutions manifest in rural solar mini-grids. Current conditions include 1–5 USD per credit, fragmented governance, and uncertain authorization. This paper asks: does carbon revenue influence mini-grid investment decisions? How do transaction costs compare to revenues at scales below 200 kW? What conditions enable successful integration at scale, including host government authorization? We address these questions through stakeholder interviews, transaction cost analysis, and a case study of the atmosfair-MiniGridCo Programme of Activities in Nigeria.

3. Conceptual Framework



Framework Logic

——→ Direct relationship/influence

.....→ Feedback loop/enabling relationship

Carbon markets provide financing → Policy creates enabling environment → Mini-grids scale up

Figure 1: Conceptual framework linking energy poverty, rural solar mini-grids, carbon markets, and policy.

The conceptual framework examines how carbon markets can support rural solar mini-grid deployment in SSA through interconnected components.

1. **Carbon markets** as a financing mechanism, where voluntary credits can improve commercial viability but face uncertainties around transaction costs and certification, host government authorization, and revenue leakage to intermediaries. [17, 18, 13]
2. **Policy, regulatory, financial and technical frameworks,** which shape enabling conditions but often lack integration between energy regulations and carbon mechanisms, particularly regarding Article 6 authorization requirements and carbon revenue treatment within tariff structures. [3, 10, 15]

Feedback loops link these components: regulatory clarity facilitates carbon market participation, and carbon finance strengthens commercial viability, enabling larger-scale projects. The framework guides investigation of barriers and enabling conditions across both RQs, with particular attention to structural market failures that prevent aggregation mechanisms from emerging organically.

Appendix A. Example Appendix Section

Appendix text.

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